Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/28/2017
PI Name:	Savage, Richard M.S.		
Project Title:	Multi-Purpose X-ray System		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	richardsavage@austin.rr.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	(512) 997-7780
Organization Name:	Stellarray, Inc.		
PI Address 1:	9210 Cameron Road, Suite 300		
PI Address 2:			
PI Web Page:			
City:	Austin	**State:**	TX
Zip Code:	78754-3971	**Congressional District:**	35
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	04/07/2017	**End Date:**	10/06/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
Contact Email:	jay.lemery@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Stellarray proposes the development of a highly novel Multi-Purpose X-ray Source and System (MPXS), for use on flight missions, space stations, planetary excursions, and planetary or asteroid bases, to meet nearly all NASA imaging needs as detailed in the Exploration Medical Condition List (EMCL). This proposal goes far beyond an incremental increase in imaging capability and offers a path towards providing the full range of radiographic imaging - 2D, digital tomosynthesis and even half (180 degrees) or full (360 degrees) computed tomography (CT) to cover routine and emergency imaging needs in space mission environments. The source is comprised of sections, each designed for a specific range of x-ray imaging or analysis functions. In the starting design, each section is close to one of the sidewalls of the source, which is shaped as a rectangular box and made primarily of aluminum nitride ceramic (AlN) sheets. Each AlN sidewall has a window that allows the x-ray flux to exit the source. The window can be a hollowed out section of the sidewall or a thin strip of low Z material (glass, BeO, etc.) fritted, fused, or brazed into or over a window aperture in the sidewall. Thin strips of metal can be placed over the windows for beam filtration. Each window will output flux from		

Task Description:	one or more rows of spots (x-ray pixels, or xels) on the metal anode inside. Systems comprise one or more of these sources and flat panel x-ray detectors, with several system modes configurable using the same source and detector. POTENTIAL NASA COMMERCIAL APPLICATIONS: MPXS would provide greatly enhanced imaging capabilities for the dental and limited MSK imaging of current interest to NASA. MPXS systems would go beyond the capabilities currently planned by enabling 3D/tomographic imaging, which will be particularly useful in MSK imaging and could be useful in some dental imaging. In addition, MPXS can be used for a much greater number of medical conditions of interest to NASA, particularly head and neck injuries, several conditions requiring imaging of the chest area and dual energy X-ray osteoporosis imaging. The specific gaps the proposed work addresses are 4.02 (We do not have the capability to provide non-invasive medical imaging during exploration missions) and 3.03 (We do not know which emerging technologies are suitable for in-flight screening, diagnosis, and treatment during exploration missions). Although not a focus in Phase I, MPXS can also be used for applications such as 4.27 (We do not have the capability to sterilize medical equipment during exploration missions). Sections of MPXS sources can be configured for x-ray sterilization. Other applications could include sources for a range of instruments NASA uses on space missions, including XRF and XRD. The sources carried on board for imaging applications could be used for sample analysis both on spacecraft and bases and perhaps even during excursions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	POTENTIAL NON-NASA COMMERCIAL APPLICATIONS: There are many commercial applications of the core technology to be developed in this project, including a wide range of medical and industrial imaging systems. Stellarray's smart x-ray sources can be used in various x-ray medical imaging systems. Stellarray will develop some systems on its own and sell sources for other applications where larger companies are better suited to clinical trials and market entry. MPXS sources will be sold to other developers, particularly at universities in medical schools, a number of which have asked for our resources as they are developed. They could be sold at $75K range to these developers for a good business line. By the time NASA is testing MPXS devices for missions there could be a reasonably sized installed base at universities that could also contribute application designs and reconstruction algorithms.
Task Progress:	New project for FY2017. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)